Fiscal Year:	FY 2017	**Task Last Updated:**	FY 12/12/2016
PI Name:	Derby, Jeffrey J. Ph.D.		
Project Title:	Modeling Detached Bridgman Crystal Growth		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Minneapolis	**State:**	MN
Zip Code:	55455-0132	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	01/01/2016	**End Date:**	12/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rogers, Jan	**Contact Phone:**	256.544.1081
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Volz, Martin Ph.D. (NASA Marshall Space Flight Center) Croell, Arne Ph.D. (NASA Marshall Space Flight Center)		
Grant/Contract No.:	NNX16AC37G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (February 2016): Continuation of "Modeling of Particle Transport in the Melt and Its Interaction with the Solid-Liquid Interface," grant NNX10AR70G, with the same Principal Investigator. The primary goal of this research project is to understand detached Bridgman growth in both microgravity and terrestrial environments. Detached Bridgman growth is a potentially transformative method to grow large electronic and photonic crystals. It originated from the serendipitous observation of unusual behavior exhibited by early, space-based melt crystal growth experiments, where the melt dewetted the wall, allowing the crystal to pull away and grow detached from the ampoule. The detached growth mode eliminated deleterious interactions between the growing crystal and the ampoule wall, producing crystals of heretofore unprecedented levels of structural perfection. However, what was so readily attained in space is far more difficult on Earth. There is an opportunity to continue to advance our understanding of detached Bridgman growth under terrestrial conditions by a combination of ground-based modeling coupled with microgravity and terrestrial experiments. Specifically, this project will develop and apply rigorous thermal-capillary models employing finite-element methods and coupled with furnace-scale heat transfer models to describe detached solidification. These models will be applied to assess the onset, stability, and termination of detached growth and to further ascertain the details about why this growth is so much more easily attained in microgravity environments than on Earth. Of great importance is understanding how larger-scale, Earth-based systems may be designed and operated successfully. This work will promote a collaboration between research groups at the University of Minnesota (UMN) and the NASA Marshall Space Flight Center (MSFC). A research team at MSFC will focus on experimental aspects of detached Bridgman growth, while UMN researchers will develop and apply sophisticated mathematical models to analyze the process. Experiments and theory will be conducted with complementary goals; the synergy between the two approaches will promote progress much faster than pursuing either path alone. The proposed research teams two of the foremost crystal growth groups in the world to leverage their strongly complementary skills toward the realization of the great potential of detached solidification.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Enabling the detached Bridgman process promises a means to produce large, single crystals of compound semiconductors with levels of structural perfection heretofore unattainable. This will advance the quality and affordability of cadmium telluride and cadmium zinc telluride, materials needed for high-sensitivity radiation detectors that are of great strategic importance for nuclear nonproliferation applications and securing national borders. Detached Bridgman growth may also promote significant quality increases for gallium arsenide and indium phosphide, benefiting the continued development of solid-state lighting and high-speed integrated circuits thus impacting energy conservation and electronics systems development.
Task Progress:	This year's work included finishing the last studies and papers on the prior project on particle engulfment. Unfortunately, we were not able to attract a first-year graduate student to the new project on detached Bridgman crystal growth from the incoming class of Fall 2016. Nevertheless, Principal Investigator Derby visited Dr. Martin Volz at the NASA Marshall Space Flight Center on November 29, 2016, to discuss past progress and future research directions. While work on modeling detached Bridgman crystal growth will be delayed, we expect to be able to find a student sometime in the next calendar year.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	Tao Y, Derby JJ. "Analysis of Particle Engulfment Dynamics during Solidification." TMS 2016. 145th Annual Meeting, The Minerals, Metals and Materials Society, Nashville, TN, February 14-18, 2016. TMS 2016. 145th Annual Meeting, The Minerals, Metals and Materials Society, Nashville, TN, February 14-18, 2016. Conference program. , Feb-2016
Abstracts for Journals and Proceedings	Friedrich J, Jauss T, Croell A, Sorgenfrei T, Reimann C, Tao Y, Derby JJ. "Particle incorporation during solidification of silicon under microgravity." ICCGE-18, the 18th International Conference on Crystal Growth and Epitaxy, Nagoya, Japan, August 7-12, 2016. ICCGE-18, the 18th International Conference on Crystal Growth and Epitaxy, Nagoya, Japan, August 7-12, 2016. Conference program. , Aug-2016
Abstracts for Journals and Proceedings	Tap Y, Derby JJ. "A fundamental analysis of particle engulfment dynamics during crystal growth." ICCGE-18, the 18th International Conference on Crystal Growth and Epitaxy, Nagoya, Japan, August 7-12, 2016. ICCGE-18, the 18th International Conference on Crystal Growth and Epitaxy, Nagoya, Japan, August 7-12, 2016. Conference program. , Aug-2016